

Zoe Thompson

Embedded Systems & IoT Firmware Engineer • 3 Years Experience • MERIQ Index: 72% (Developing)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Python, I2C/SPI/UART, MQTT, FreeRTOS, PCB Debugging, Embedded Linux

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *Intel* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Python service, reducing latency by 51%.
- Led a squad of 3 engineers delivering scalable microservices processing 40M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 32%.

Associate Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 96% SLA uptime.
- Optimized database indexing and caching layers using MQTT saving \$22k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Python, I2C/SPI/UART, MQTT, FreeRTOS]

Engineered end-to-end production solution leveraging Python and I2C/SPI/UART with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [MQTT, FreeRTOS, PCB Debugging, Embedded Linux]

Architected high-resilience system integrating MQTT with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Computer Science

IIIT Hyderabad — 2019 - 2023 (Score: 3.51 / 4.0)